Lab 16 Worksheet — CPU Architecture, Sockets and Cooling Solutions

Name: ____ **Date:** __

Exam objective: Compare CPU architectures, socket types and cooling solutions, and diagnose thermal problems from observed behaviour (Core 1 objectives 3.4 and 5.2).

Record as you go

Step	What you did	What you observed
1	Compare x86-64 and ARM on power consumption, heat output, software compatibility and typical device class, and record which dominates desktops and which dominates mobile.	
2	Record the socket types: LGA with pins on the motherboard used by Intel, PGA with pins on the processor still used by some AMD parts, and BGA soldered permanently in mobile devices.	
3	Explain why a BGA processor in a laptop cannot be upgraded, and what that means when advising a customer on a slow machine.	
4	Define core count, thread count and simultaneous multithreading, and explain why an eight-core processor may report sixteen logical processors.	
5	Open Task Manager, go to Performance then CPU, and record the reported cores, logical processors, base speed, current speed and virtualization state.	
6	Explain thermal throttling: a processor reduces its clock speed when it exceeds its temperature limit, which protects the silicon but degrades performance.	

Step	What you did	What you observed
7	Explain the difference between throttling and overclocking, and note that Intel Turbo Boost and AMD Precision Boost are sanctioned automatic overclocking within thermal headroom.	
8	Compare passive cooling — heatsink, thermal paste and heat pipes with no moving parts — against active cooling with a fan, and note where each is appropriate.	
9	Explain the role of thermal paste: it fills the microscopic gaps between the processor's heat spreader and the cooler base so heat can actually transfer, and record that too much is as bad as too little.	
10	Compare air cooling against liquid cooling on cooling capacity, noise, cost and failure mode, noting that a liquid cooler leak can destroy the whole system.	
11	Build the thermal fault diagnostic sequence: confirm all fans spin, check intakes and heatsink fins for dust, verify the cooler is properly seated, check paste condition, then check ambient temperature and case airflow.	
12	Monitor processor temperature under load and record the idle temperature, the load temperature and whether the clock speed dropped, then state whether the cooling is adequate.	

Verification

Success criterion: Task Manager confirms the core and logical processor counts and the virtualization state; your diagnostic sequence orders the checks from cheapest to most invasive; and the temperature record supports a stated conclusion about cooling adequacy.

- I completed every step in the lab.
- My result meets the success criterion above.

- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
